PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Online Synchronous Chat Counselling for Young People Aged 12-25: A Mixed Methods Systematic Review Protocol.
AUTHORS	Tibbs, Maria; O'Reilly, Aileen; Dwan O'Reilly, Maeve; Fitzgerald, Amanda

VERSION 1 – REVIEW

REVIEWER	Edirippulige, S University of Queensland
REVIEW RETURNED	30-Jan-2022

GENERAL COMMENTS	Thank you for the opportunity to review this paper/review protocol. I think the authors have made a good effort addressing most of the requirements relating to the study/review. I would recommend the authors to briefly talk about the 'other online methods' in the discussion especially asynchronous chat counselling as this directly contrast to the focus of the authors. That may help authors to highlight the rationale of their choice. They may add few relevant citations such as: Navarro P, Bambling M, Sheffield J, Edirippulige S Exploring Young People's Perceptions of the Effectiveness of Text-Based Online Counseling: Mixed Methods Pilot Study, JMIR Ment Health 2019;6(7):e13152 URL: https://mental.jmir.org/2019/7/e13152; DOI: 10.2196/13152 Navarro P, Bambling M, Sheffield J, Edirippulige S. Exploring Young People's Perceptions of the Effectiveness of Text-Based Online Counseling: Mixed Methods Pilot Study. JMIR Ment Health. 2019 Jul 3;6(7):e13152. doi: 10.2196/13152. PMID: 31271149; PMCID: PMC6636236. Apart from that I think the review is in good shape for the acceptance.

REVIEWER	Pretorius, Claudette University College Dublin, Computer Science
REVIEW RETURNED	10-Feb-2022

GENERAL COMMENTS	This protocol outlines a mixed-methods systematic review examining the effectiveness online synchronous chat interventions. Strengths of this study include it's mixed-methods design, the consultation with young people on the research questions and research design, focusing specifically on synchronous online

	counselling as well as having a comprehensive analysis plan in place. I do have the following questions and concerns. Including papers from as far back 1995 might not be helpful or relevant as the technology and mediums of delivering online counselling have vastly changed in the last 30 years. It is likely that studies prior to 2000 or even 2005, will speak to very different, and very likely very irrelevant issues than those faced by young people today. Most ordinary households did not have access to the Internet in 1995 and so I wonder about the relevance or utility of including studies as far back as this. Secondly, greater detail on how quantitive data will be transformed and included would also be beneficial. Thirdly, be clear on the definitions of terms such as acceptance and acceptability which have very specific meanings and uses. Please see 'Technology acceptability, acceptance and adoption - definitions and measurement' by Nadal, Sas & Doherty (2019). The findings from this review will provide valuable knowledge in understanding the processes and experiences of those young people who have engaged in synchronous online counselling. I look forward to the results.

VERSION 1 – AUTHOR RESPONSE

Reviewers' Comments to the Author:
Reviewer 1

Comments to the Author:
Thank you for the opportunity to review this paper/review protocol. I think the authors have made a good effort addressing most of the requirements relating to the study/review.
Author response:
Thank you!

1. I would recommend the authors to briefly talk about the 'other online methods' in the discussion especially asynchronous chat counselling as this directly contrast to the focus of the authors. That may help authors to highlight the rationale of their choice. They may add few relevant citations such as:
   Navarro P, Bambling M, Sheffield J, Edirippulige S
   Exploring Young People's Perceptions of the Effectiveness of Text-Based Online Counseling: Mixed Methods Pilot Study, JMIR Ment Health 2019;6(7):e13152
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   Navarro P, Bambling M, Sheffield J, Edirippulige S. Exploring Young People's Perceptions of the Effectiveness of Text-Based Online Counseling: Mixed Methods Pilot Study.
   JMIR Ment Health. 2019 Jul 3;6(7):e13152. doi: 10.2196/13152. PMID:31271149; PMCID: PMC6636236.

Author response:
The authors appreciate that highlighting the distinction between asynchronous and synchronous chat could provide a stronger rationale for choice. As such, the below was included in the introductory section of this paper (pg. 5-6, lines 103-107).

The instant, real-time nature of online synchronous chats is of distinct benefit when compared to other modalities such as email-based asynchronous chat. This is evidenced in recent literature where young people and mental health professionals both highlighted 'poor timeliness and response' as a disadvantage of their email and text services [22 23].

Reviewer 2

1. Including papers from as far back 1995 might not be helpful or relevant as the technology and mediums of delivering online counselling have vastly changed in the last 30 years. It is likely that studies prior to 2000 or even 2005, will speak to very different, and very likely very irrelevant issues than those faced by young people today. Most ordinary households did not have access to the Internet in 1995 and so I wonder about the relevance or utility of including studies as far back as this.

Author response:

Thank you for your feedback. Yes, papers included as far back as 1995 may indeed lack some relevance. This timeframe was initially chosen as the first occurrence of text-based online support was launched in 1995. However, the authors can confirm that following initial abstract and title screening, the earliest included study was published in 2002. Most studies included following title and abstract screening were published in the decade spanning from 2010-2020.

1. Secondly, greater detail on how quantitative data will be transformed and included would also be beneficial.

Author response:

Quantitative findings will be transformed, and a textual description of findings will be provided. This will allow for descriptive vote-counting of the direction of effects. Further, adopting guidance provided by Sandelowski (2000), study samples and research results, for example, details of the intervention, participants and outcomes will be described textually to allow for the description of most salient or commonly occurring attributes. To assist in this textual description, the TIDieR framework will be used. The following was added to the study design and analysis section (pg. 10, lines 207-213).

A mixed methods review adopting a data-based convergent integrated design will be employed. This will be informed by the Joanna Briggs Institute methodology[45], whereby quantitative data will be transformed and a textual description of study characteristics (sample size and demographics, intervention description, processes employed, acceptance and effectiveness) and findings will be provided. This will be combined with qualitative findings using the same categorisation. This will be guided by the TIDieR framework (see below for detailed description). Addressing effectiveness, effect direction will be used, that is, evidence for improvement, deterioration, or no evidence of change within outcome measurement. This method is commonly adopted in reviews with heterogeneous outcomes as well as those employing diverse effect measurement [46].

1. Thirdly, be clear on the definitions of terms such as acceptance and acceptability which have very specific meanings and uses. Please see 'Technology acceptability, acceptance and adoption -definitions and measurement' by Nadal, Sas & Doherty (2019).

Author response:

Thank you for this very useful paper. The authors are sensitive to the multi-dimensional and complex nature of 'acceptability' and the resulting difficulties with defining this concept. The authors had initially used a definition provided by Sekhon, Cartwright & Francis (2017) and anticipated examining 'retrospective acceptability'. However, upon examining the paper kindly provided in addition to follow-up literature such as 'Acceptance and acceptability criteria: a literature review by Alexandre et al., 2018', terminology within the protocol has been amended. Authors will now adopt the term 'acceptance' to capture users' perceptions of the intervention after use. As such a clearer description of what is meant by acceptance has been added (pg. 6, lines 115-116) See below. Additionally, the term acceptability has been changed to 'acceptance' throughout.

Moreover, findings examining the acceptance of these interventions appear to be limited, that is studies focusing on the user experiences, perceptions, and satisfaction post engagement [26 27].

The findings from this review will provide valuable knowledge in understanding the processes and experiences of those young people who have engaged in synchronous online counselling. I look forward to the results.
Author response:
Thank you!

Additional references

Navarro P, Sheffield J, Edirippulige S, et al. Exploring Mental Health Professionals' Perspectives of Text-Based Online Counseling Effectiveness With Young People: Mixed Methods Pilot Study. Jmir Mental Health 2020;7(1) doi: 10.2196/15564

Navarro P, Bambling M, Sheffield J, et al. Exploring Young People's Perceptions of the Effectiveness of Text-Based Online Counseling: Mixed Methods Pilot Study. JMIR mental health 2019;6(7):1. doi: http://dx.doi.org/10.2196/13152

Nadal C, Sas C, Doherty G. Technology acceptance in mobile health: scoping review of definitions, models, and measurement. Journal of Medical Internet Research 2020;22(7):e17256.

Alexandre B, Reynaud E, Osiurak F, et al. Acceptance and acceptability criteria: a literature review. Cognition, Technology & Work 2018;20(2):165-77.

Boon MH, Thomson H. The effect direction plot revisited: Application of the 2019 Cochrane Handbook guidance on alternative synthesis methods. Research Synthesis Methods 2021;12(1):29-33. doi: https://doi.org/10.1002/jrsm.1458

VERSION 2 – REVIEW

REVIEWER	Pretorius, Claudette University College Dublin, Computer Science
REVIEW RETURNED	22-Mar-2022

GENERAL COMMENTS	Thank you to the authors for their detailed responses and for clarifying concerns.